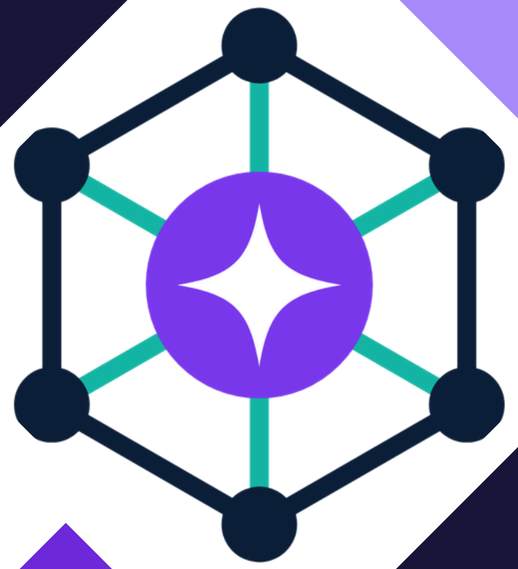


Self Network Deployer

AI-DRIVEN AUTOMATED DEPLOYMENT OF 5G TEST ENVIRONMENTS

M3: Digital Accessibility and Usability

instituto de telecomunicações
universidade de aveiro
ATNOG

Meet the Team

Projeto em Engenharia Informática · Universidade de Aveiro



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Recap

Deploy a 5G network by asking for it

1

You ask

"I want a 5G network with 10 phones and 2 slices"

2

The AI proposes a configuration

MCC · TAC · DNN · UEs · slices

3

You review and confirm

Nothing is deployed until a person says yes.

4

SND deploys and tests it

Open5GS + UERANSIM

✓ Network deployed

The user has to read, understand and confirm every step,
so every step has to be accessible.

RGPD

Data collected	Purpose / Rationale	Anonymisation / Pseudonymisation	Right to erasure	Privacy Policy	Security Measures
Username	User identification and display.	Link data through internal identifiers to avoid unnecessary direct identification.	Deleted when the account is deleted.	No sharing with external third parties is planned.	Access control, HTTPS and restricted PostgreSQL access.
Email and access role	Identify the account, authenticate users and distinguish administrators.	Use internal identifiers where email is not needed.	Deleted with the account.	No external sharing planned. Institutional LDAP authentication may be added.	HTTPS and role-based access control.
Password (hash)	Secure credential verification during authentication.	Not applicable as a separate technique. Hashing protects passwords but is not anonymisation.	Hash deleted with the account.	Not shared with third parties or AI agents. LDAP may remove the need for local passwords.	Argon2id; minimum 8 characters; limits on login attempts.
System interaction history	Review past requests, continue conversations and refine configurations.	Use user_id and minimise personal data sent to AI agents.	Users may delete conversations and associated history.	Only necessary content may go to the institutional AI server; no other external sharing planned.	Per-user access controls, HTTPS and data minimisation.
NetworkSpec and 5G configurations	Save, view, edit, deploy and reuse network configurations.	Link via internal identifiers; avoid unnecessary personal data.	Users may delete saved configurations; temporary data is removed when the network stops.	No external sharing planned; required data goes to deployment infrastructure.	Access control and isolation between users.
System logs	Detect failures, diagnose issues, support security and audit operations.	Use internal identifiers where possible; exclude unnecessary personal data.	Deleted after the retention period (to be defined).	Access restricted to authorised system administrators.	Do not log passwords or full tokens; access control and defined retention.

RGPD

Data collected

1. Name
2. Email and access role
3. NetworkSpec and 5G configurations

Purpose / Rationale

1. Identify the user
2. Identify the account and permissions
3. Reuse previous configurations

Anonymisation / Pseudonymisation

1. Internal identifiers (user_id)
2. Internal identifiers (user_id)
3. Internal identifiers (user_id)

Right to erasure

1. Deletion of the account and associated data
2. Deletion of the account and associated data
3. Delete conversations and saved networks

Privacy Policy

1. No external sharing is planned
2. No external sharing is planned
3. Required data will be sent to the deployment infrastructure, but never personal data.

Security Measures

1. HTTPS and access control
2. HTTPS and role- and permission-based access control
3. Access control and isolation between users

Software Licensing

Third-party software: license type and implications

Technology	License	Type & implications
React	MIT	Permissive. Use, modify and sell; retain copyright and license notices.
FastAPI	MIT	Same rights and obligations as React.
NGINX	BSD-2-Clause	Permissive, with copyright notice requirements.
PostgreSQL	PostgreSQL License	Permissive, including modification and redistribution with notices.
Python	PSF License 2	Permissive, with attribution and change notices for redistribution.
Jinja2	BSD-3-Clause	Permissive; no endorsement using original authors' names.
Docker Engine	Apache 2.0	Permissive, includes an express patent grant.
NuExtract3	Apache 2.0	Permissive AI model; reuse commercially with required notices.
Ansible	GPL-3.0-or-later	Copyleft; distribution of covered combined/modified works may require corresponding source.
Proxmox VE	Mainly AGPL-3.0	Copyleft; verify licenses for individual components.
Open5GS	AGPL-3.0	Network copyleft; modified versions may require source access for remote users.
UERANSIM	AGPL-3.0 / commercial	Dual license; commercial agreement is an option when AGPL is unsuitable.
iperf3	BSD-3-Clause	Permissive; commercial use and redistribution allowed with notices.
Qwen3.8-Flash-Next	Qwen Community License 1.0	Custom license; commercial use has specific conditions.
Qwen3-Embedding-4B	Apache 2.0	Permissive; commercial use and modification allowed with required notices.

Intellectual Property & Compliance

GPL / AGPL

Check obligations if covered code is modified or redistributed; AGPL may require source access for network users of modified versions.

AI model terms

Verify NuExtract version and review Qwen3.8-Flash-Next Community License 1.0, including commercial-service conditions.

RAG documentation

Verify reuse permissions for external manuals, API documentation and examples added to the knowledge base.

Compliance checklist

Track versions and licenses, keep copyright notices, document modifications and distribution.

API integration does not automatically make SND's separate code GPL/AGPL

Terms & Conditions

Key conditions governing the use of SND

Conditions of Use

- Authorized research use only.
- User approval required before deployment.
- Access restricted to own networks.

Liability Disclaimers

- AI-generated configurations may contain errors.
- Successful deployment is not guaranteed.
- Not intended for production networks.

Refund & Cancellation Policy

- No paid services or subscriptions planned.
- Commercial refund policies do not apply.

Legal Jurisdiction

- Applicable Portuguese and EU legislation.
- Relevant institutional policies.

Changes to Terms

- Updates may reflect platform or legal changes.
- Users should be informed of significant changes.

Sector-specific Regulations

Navigating telecommunications law, global 5G standards, and AI safety

TELECOM LAW

ANACOM Exemption

- **Legal Context:** Under Portuguese Law no. 16/2022, public operators require ANACOM licenses & radio spectrum
- **R&D Exemption:** Designed for private testbeds across academic institutions, R&D labs, and private environments (e.g., IT).
- **No Public Broadcast:** Operates in containerized environments (Docker/VMs) with simulated radio (UERANSIM).
- **Verdict:** Legally exempt from commercial operator obligations and spectrum licensing.

STANDARDS

3GPP & ETSI Compliance

- **3GPP Releases 15-17:** Network blueprints (NetworkSpec) follow the official 5G Core Service-Based Architecture (SBA).
- **Core Network Functions:** Native alignment with standard 5G entities: AMF, SMF, and UPF.
- **Standard Identifiers:** Configures official ITU/3GPP test network codes (PLMN 001-01).
- **ETSI NFV Alignment:** Decouples software from hardware, enabling seamless future connection to real gNodeBs.

AI GOVERNANCE

EU AI Act Governance

- **Risk Categorization:** Classified as minimal / limited risk technical assistant for network engineers.
- **Human-in-the-Loop:** AI plans, reviews, and suggests parameters, but NEVER executes autonomously.
- **Mandatory Approval:** Deployments, rollbacks, and slice configs require explicit user confirmation.
- **Fail-Safe Operation:** Mitigates hallucinations or configuration drifts before impacting testbed resources.

Who can't use SND?

Users

ATNOG researchers (main)
Students (future)

Reference

WCAG 2.2, Level AA

Profiles analysed

Blind · Low vision · Colour blindness · Motor impairments · Cognitive / low digital literacy

Not applicable

Deaf users: SND has no audio or video

How we audited

Contrast (WebAIM) · axe DevTools · Keyboard only · Colour-blindness simulation · 200% zoom · Screen reader (Orca)

What we found

20

problems found

each one linked to a WCAG rule

19

fixed

axe DevTools: 0 issues on all 8 screens

By severity

- 3 High
- 10 Medium
- 7 Low

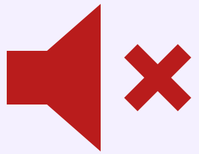
Already good

- ✓ Main task works with keyboard only
- ✓ No information relies on colour alone
- ✓ Forms have real labels and text error messages

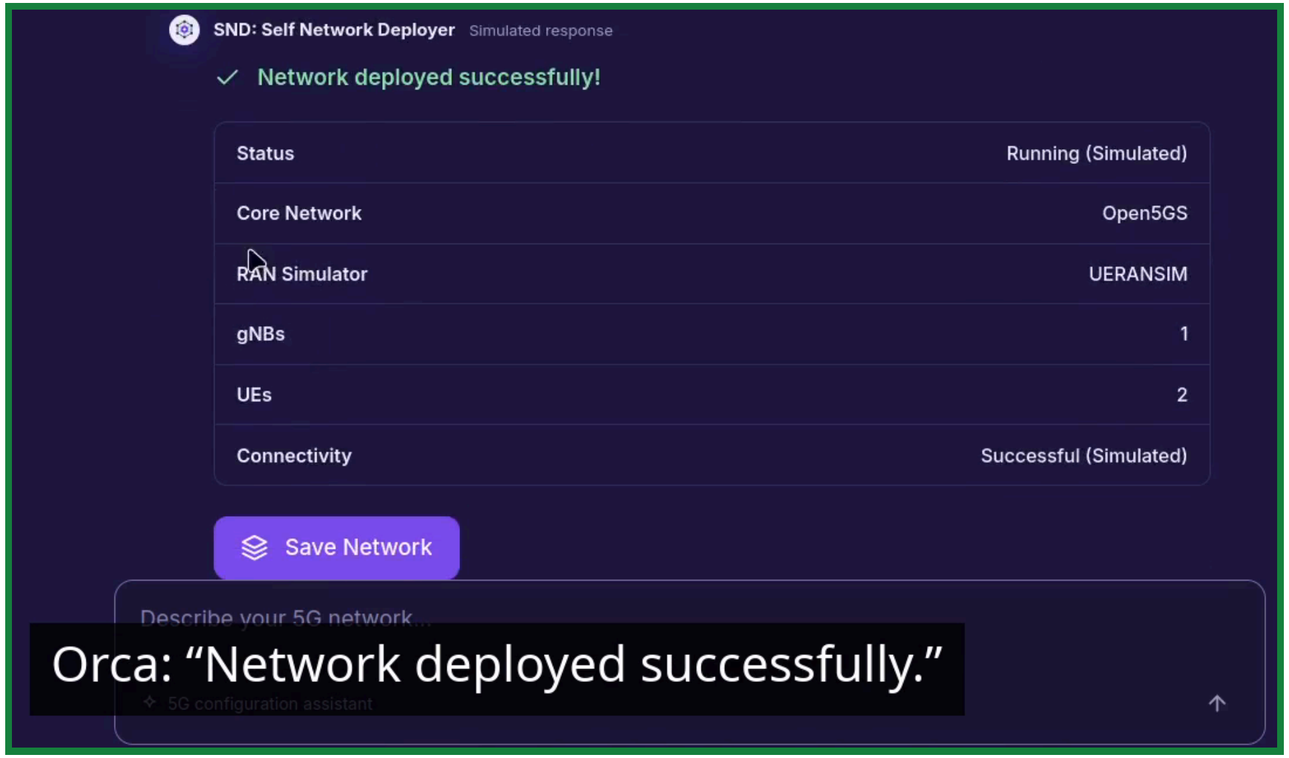
For a blind user, SND was a chat that never replied

Before

The proposal arrives → the screen reader stays silent. Same during the deploy.



After

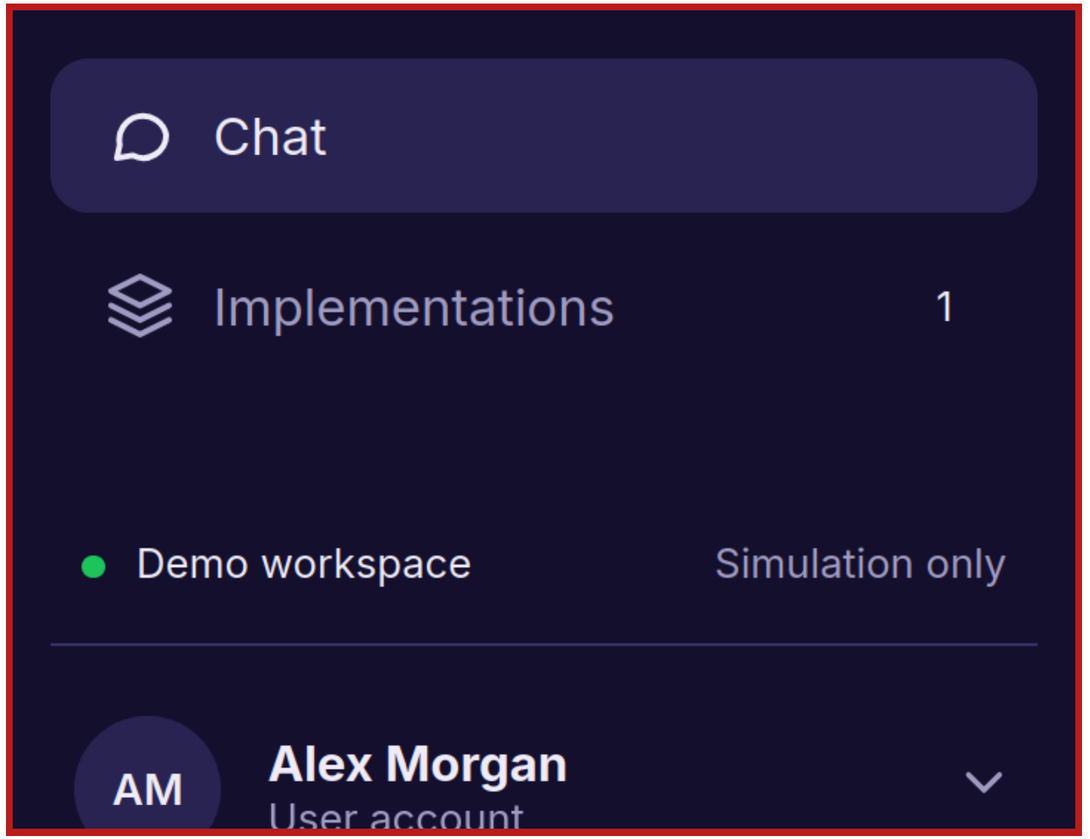


The reader announces the reply, every deploy step and “Network deployed successfully.”

WCAG 4.1.3 Status Messages · High severity

With zoom, conversations disappeared

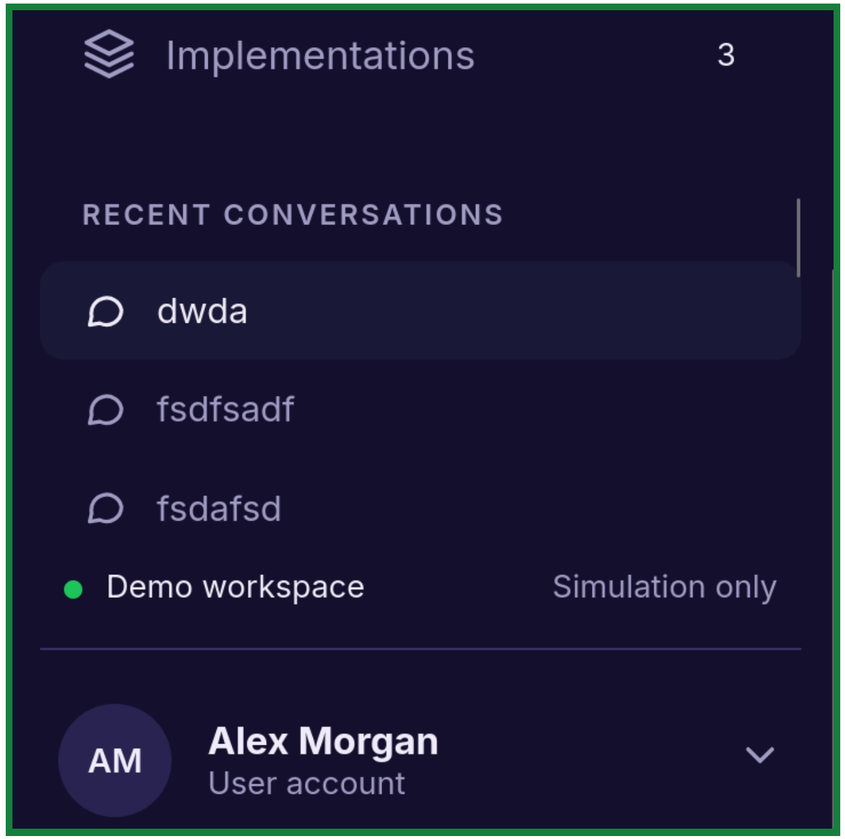
Before The conversation list disappears and Log out is cut off



WCAG 1.4.4 / 1.4.10 · High severity

With zoom, conversations disappeared

After The sidebar scrolls, so conversations are always reachable

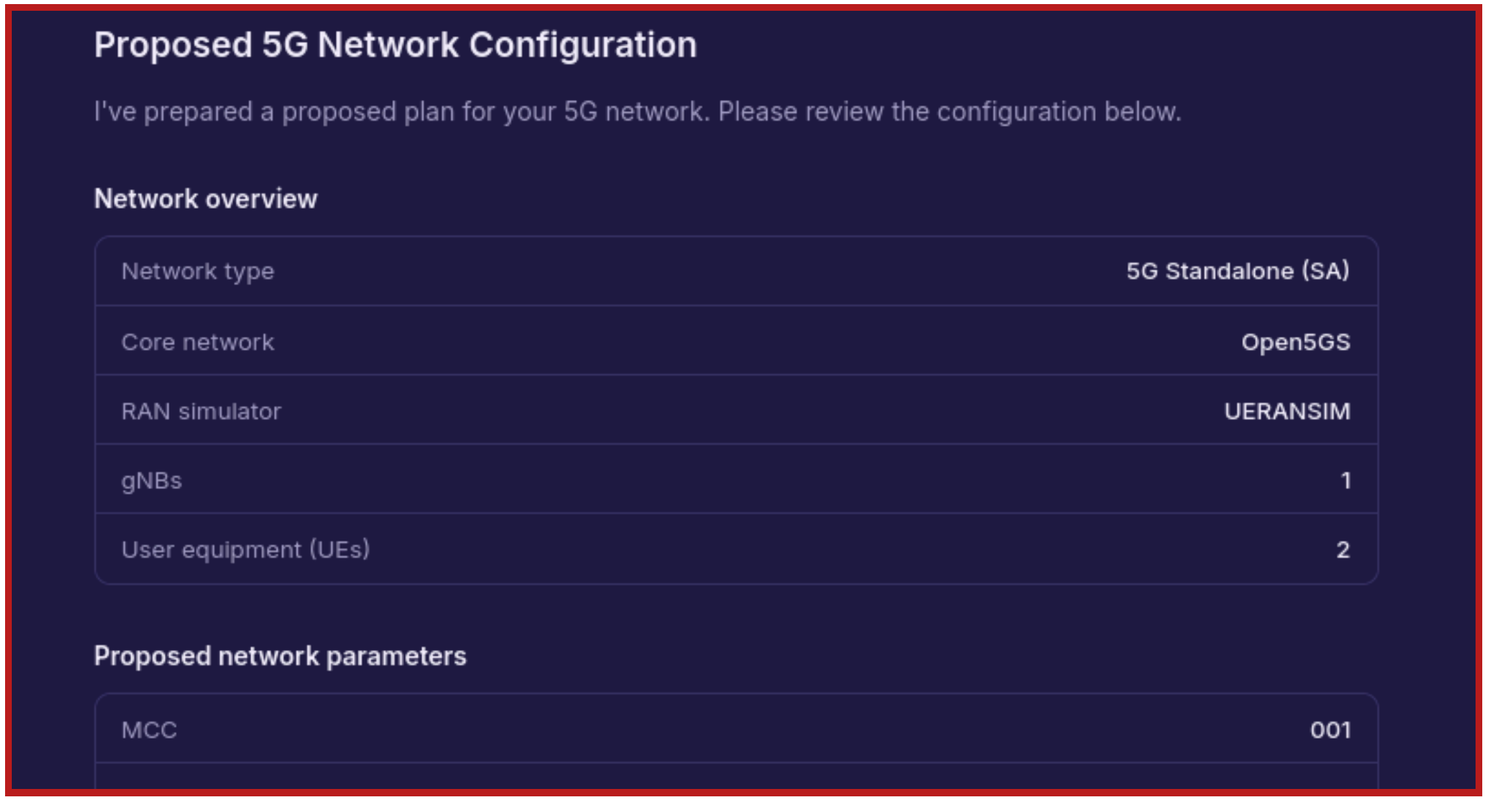


WCAG 1.4.4 / 1.4.10 · High severity

MCC, TAC, DNN… what is this?

The proposal users must confirm before deploying was full of acronyms.

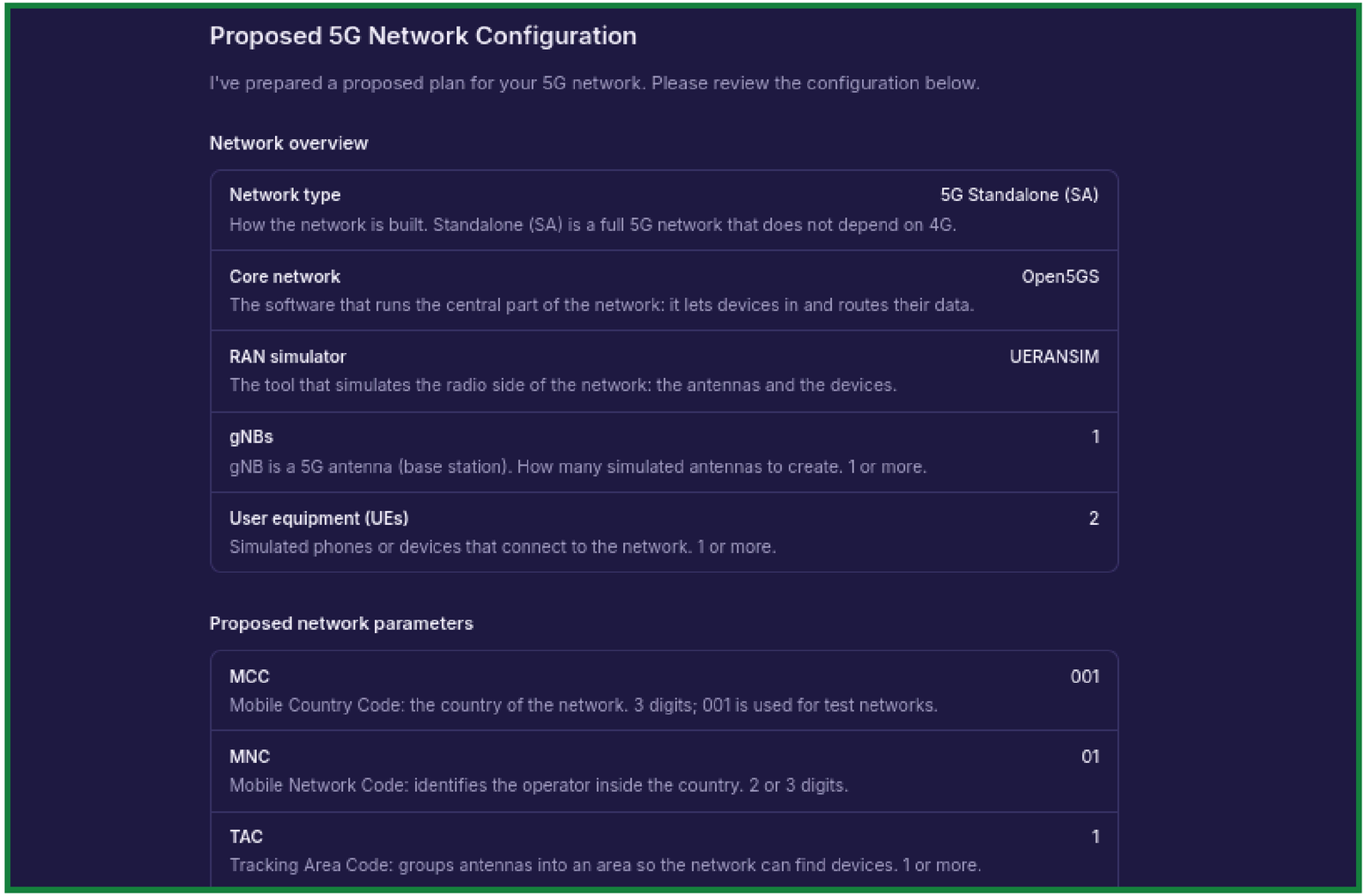
Before 5G acronyms with no explanation



WCAG 2.4.6 / 3.3.2 · Medium (High for students)

MCC, TAC, DNN… what is this?

After Every parameter explained in plain language



Proposed 5G Network Configuration

I've prepared a proposed plan for your 5G network. Please review the configuration below.

Network overview

Network type How the network is built. Standalone (SA) is a full 5G network that does not depend on 4G.	5G Standalone (SA)
Core network The software that runs the central part of the network: it lets devices in and routes their data.	Open5GS
RAN simulator The tool that simulates the radio side of the network: the antennas and the devices.	UERANSIM
gNBs gNB is a 5G antenna (base station). How many simulated antennas to create. 1 or more.	1
User equipment (UEs) Simulated phones or devices that connect to the network. 1 or more.	2

Proposed network parameters

MCC Mobile Country Code: the country of the network. 3 digits; 001 is used for test networks.	001
MNC Mobile Network Code: identifies the operator inside the country. 2 or 3 digits.	01
TAC Tracking Area Code: groups antennas into an area so the network can find devices. 1 or more.	1

WCAG 2.4.6 / 3.3.2 · Medium (High for students)

Questions?

Learn more about SND

Scan the QR code to visit our project microsite.

mariosantoslp.github.io/SND-microsite

Thank you!